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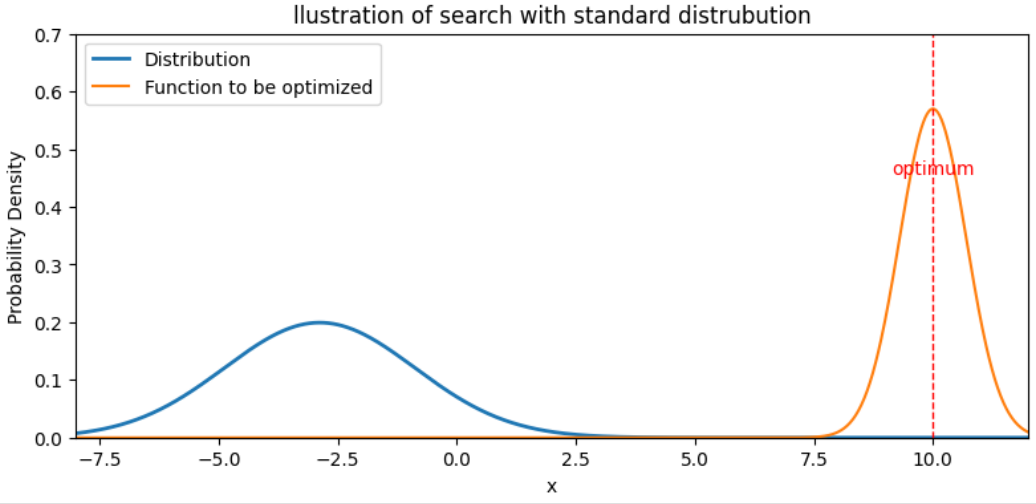



# Natural Evolution Strategies

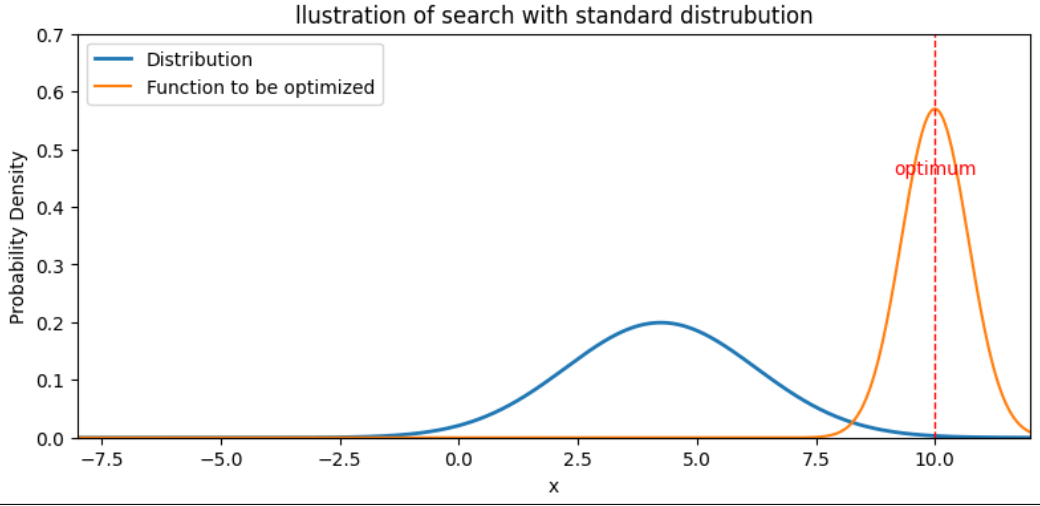
Using modified stochastic gradient on distribution optimizations.

Christof Stein, Felix Möller

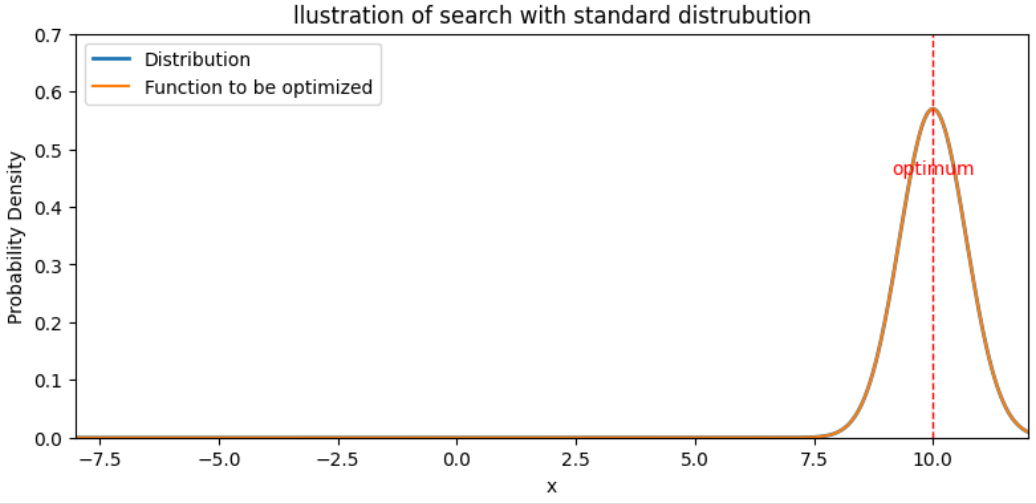
# Introduction I



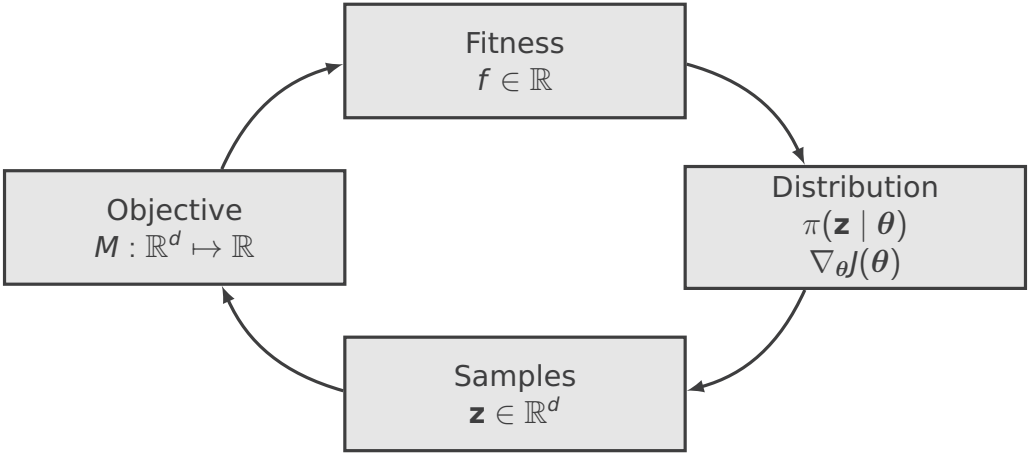
# Introduction I



# Introduction I



# Introduction II

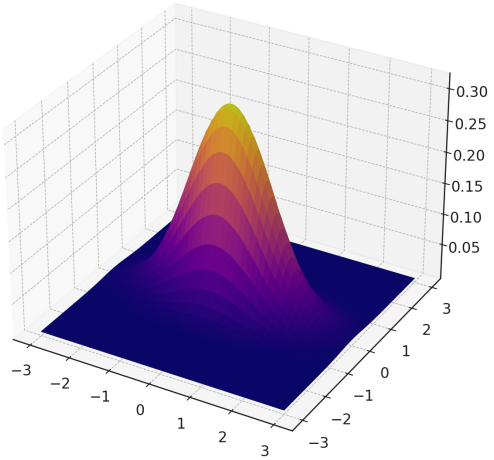


# Parameterizations I

- Parameters $\boldsymbol{\theta}$ for distribution:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_n)^T$$

- Typically:
  - Mean vector $\boldsymbol{\mu} \in \mathbb{R}^d$
  - Covariance $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$



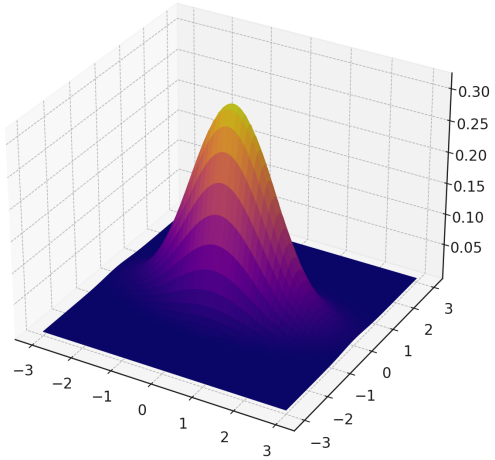
# Parameterizations II

2D Separable Normal Distribution

$$\pi(\mathbf{z} \mid \boldsymbol{\theta}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$$

Parameters:

$$\boldsymbol{\theta} = (\mu_1, \mu_2, \sigma_1, \sigma_2)^T$$



# Parameterizations III

- Monte Carlo estimate:

$$\nabla_{\boldsymbol{\theta}} J \approx \frac{1}{\lambda} \sum_{k=1}^{\lambda} f(\mathbf{z}_k) \nabla_{\boldsymbol{\theta}} \log \pi(\mathbf{z}_k)$$

- 2D Separable Normal:

$$\nabla_{\mu_i} \log \pi = \frac{z_i - \mu_i}{\sigma_i^2},$$

$$\nabla_{\sigma_i} \log \pi = \frac{(z_i - \mu_i)^2}{\sigma_i^3} - \frac{1}{\sigma_i}$$

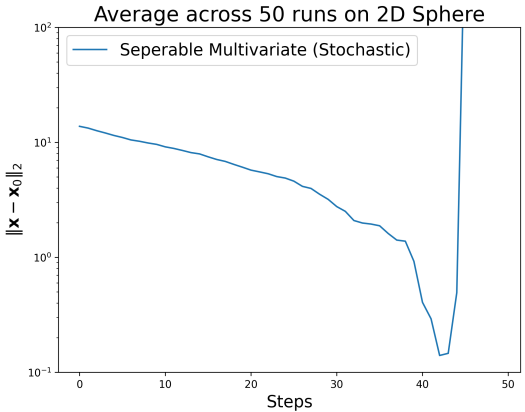
# Stochastic Gradient

- Problem:

$$\nabla_{\mu_i} J \propto \frac{1}{\sigma_i^2}$$

$$\widetilde{\nabla} J = \mathbf{F}^{-1} \nabla J$$

**Natural Gradient**



# Natural Gradient

- Fisher Information Matrix (general form):

$$\mathbf{F} = \mathbb{E}_{\mathbf{z}\sim\pi}\left[\nabla_{\boldsymbol{\theta}} \log \pi(\mathbf{z})\, \nabla_{\boldsymbol{\theta}} \log \pi(\mathbf{z})^T\right]$$

- 2D Separable Normal:

$$\mathbf{F}_{\mu_i,\mu_i} = \frac{1}{\sigma_i^2}, \quad \mathbf{F}_{\sigma_i,\sigma_i} = \frac{2}{\sigma_i^2}$$

- Corrected (natural) Gradient:

$$\widetilde{\nabla}_{\mu_i} J \propto f(\mathbf{z}_k) \cdot (z_{k,i} - \mu_i)$$

$$\mathbf{F} = \begin{bmatrix} \frac{1}{\sigma_1^2} & & & \\ & \frac{1}{\sigma_2^2} & & \\ & & \frac{2}{\sigma_1^2} & \\ & & & \frac{2}{\sigma_2^2} \end{bmatrix}$$

$$\mathbf{F}^{-1} = \begin{bmatrix} \sigma_1^2 & & & \\ & \sigma_2^2 & & \\ & & \frac{\sigma_1^2}{2} & \\ & & & \frac{\sigma_2^2}{2} \end{bmatrix}$$

# Models and Distributions I

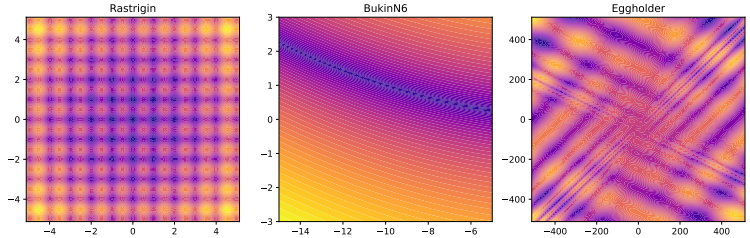
**Full Covariance Approaches:**

- **Classic Multivariate Normal**
  $\theta = (\mu, \mathbf{A})^T$
  where $^\circ = \mathbf{A}^T\mathbf{A}$ (Cholesky)
- **Exponential Multivariate Normal (xNES)**
  $\theta = (\mu, \mathbf{A})^T$
  with **A** updated via matrix exponential

**Separable Approaches:**

- **Classic Separable Normal (SNES)**
  $\theta = (\mu_1, \ldots, \mu_d, \sigma_1^2, \ldots, \sigma_d^2)^T$
- **Exponential Separable Normal**
  $\theta = (\mu_1, \ldots, \mu_d, \ln \sigma_1, \ldots, \ln \sigma_d)^T$
- **Student's-T Separable**
  $\theta = (\mu_1, \ldots, \mu_d, \ln \sigma_1, \ldots, \ln \sigma_d, \ln \nu)^T$

# Models and Distributions II




```python
x = NPolynome(np.array([[0, 0], [1, 0]]))
y = NPolynome(np.array([[0, 0], [0, 1]]))

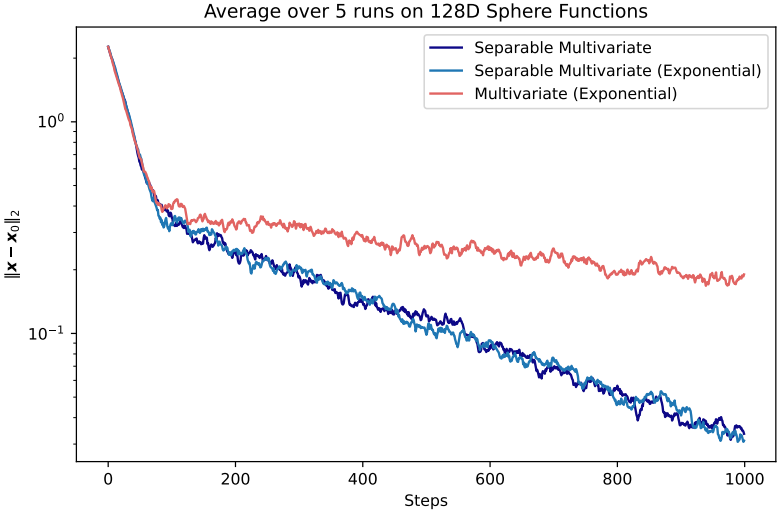
# Rastrigin
model = 10*2+x**2-10*np.cos(2*np.pi*x)+y**2-10*np.cos(2*np.pi*y)

# Buckin
model = 100*(np.sqrt(np.abs(y-0.01*x**2)))+0.01*np.abs(x+10)
```

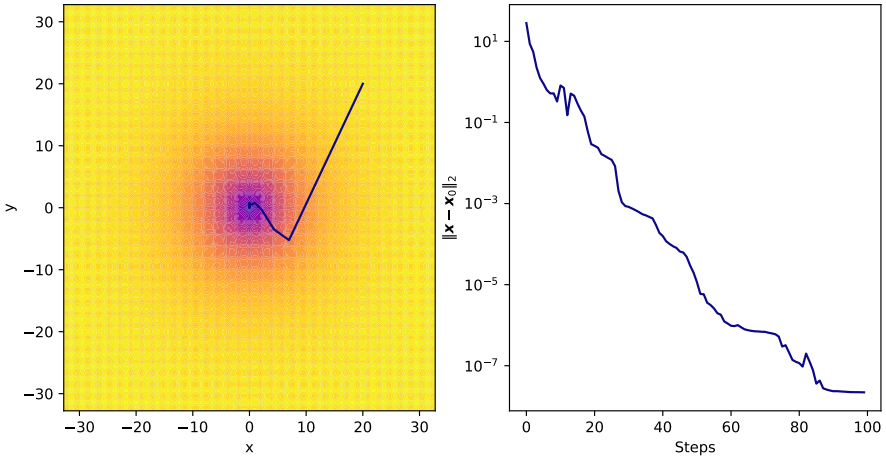
# Sphere Function I



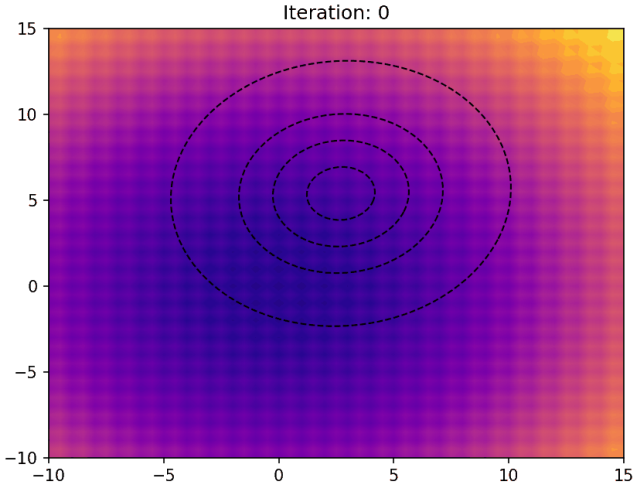
# Sphere Function II

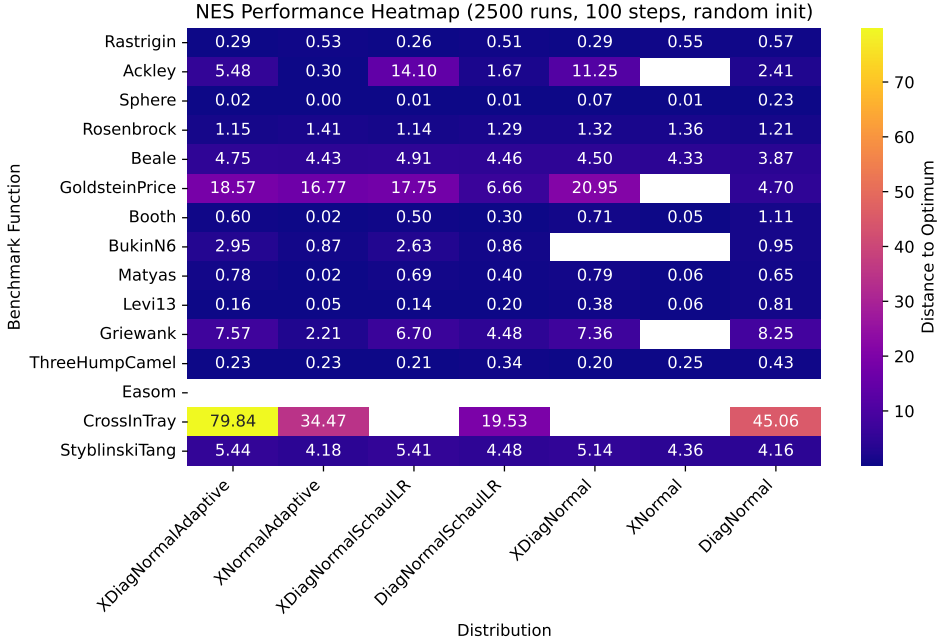


# Other Analysis



# Other Analysis




NES Performance Heatmap (2500 runs, 100 steps, random init)
Benchmark Function
Rastrigin 0.29 0.53 0.26 0.51 0.29 0.55 0.57
Ackley 5.48 0.30 14.10 1.67 11.25 2.41
Sphere 0.02 0.00 0.01 0.01 0.07 0.01 0.23
Rosenbrock 1.15 1.41 1.14 1.29 1.32 1.36 1.21
Beale 4.75 4.43 4.91 4.46 4.50 4.33 3.87
GoldsteinPrice 18.57 16.77 17.75 6.66 20.95 4.70
Booth 0.60 0.02 0.50 0.30 0.71 0.05 1.11
BukinN6 2.95 0.87 2.63 0.86 0.95
Matyas 0.78 0.02 0.69 0.40 0.79 0.06 0.65
Levi13 0.16 0.05 0.14 0.20 0.38 0.06 0.81
Griewank 7.57 2.21 6.70 4.48 7.36 8.25
ThreeHumpCamel 0.23 0.23 0.21 0.34 0.20 0.25 0.43
Easom
CrossInTray 79.84 34.47 19.53 45.06
StyblinskiTang 5.44 4.18 5.41 4.48 5.14 4.36 4.16
XDiagNormalAdaptive
XNormalAdaptive
XDiagNormalSchaulLR
DiagNormalSchaulLR
XDiagNormal
XNormal
DiagNormal
Distribution
Distance to Optimum
10
20
30
40
50
60
70